SAQI — Settings Tuning Guide

How to read, adjust and test every value in the settings UI.

Where. Open the settings page in a browser on the same network as the robot:
http://<robot-ip>:8000/settings/ui (same host/port as the camera feed).

When changes apply. Editable values are read when you switch the robot into **automatic** mode. So the flow is: edit a value, press **Save**, then turn auto OFF and ON again (Stop/Start). A change made mid-run does not disturb the run in progress; the next Start picks it up.

Save / Restore defaults. Save writes only the values you changed to a small machine-local file (runtime_settings.json). **Restore defaults** wipes those overrides so every value returns to the factory default shown in this guide.

Read-only group. The **Hardware / wiring** section is shown for reference only. Those values (GPIO pins, camera URL, serial port, etc.) are read once when the backend boots; to change them you edit components/config.py and restart the backend.

Safety. Out-of-range numbers are automatically clamped to the min/max listed here, and a bad settings file is ignored (defaults win), so you can experiment freely.

Each entry below lists the on-screen label, the internal key, its default and allowed range, then **what it does** and **how to tune it**.

Mission / Return to base

How many plants to water before driving home, and how the base QR code is recognised.

Plants per run (PLANTS_PER_RUN)

Default: **2** | Range: 0 to 100 | whole number

What it does: The number of watering events that complete before the robot stops hunting for plants and drives home to the base QR code. It counts how many times the pump runs on a plant, not distinct plants — so if the same plant is watered twice it counts as two.

Tuning: Set it to the number of plants in the row you want watered. Use 0 for 'unlimited' — the robot waters forever and never returns to base (the original behaviour). To quickly test return-to-base, set it to 1: the robot waters one plant, logs 'all plants watered', and immediately starts looking for the base.

Base QR text (BASE_QR_PAYLOAD)

Default: **SAQI_BASE** | text

What it does: The exact text the base QR code must decode to. Any other QR in view is ignored. This is how the robot tells 'home' apart from a random sticker.

Tuning: Must match the text encoded in the QR you printed, character-for-character and case-sensitive (default 'SAQI_BASE'). Only change it if you reprint the QR with different text. No spaces or URLs — keep it short so the QR stays low-density and easy to read from a distance.

Base arrival size (BASE_ARRIVAL_AREA_RATIO)

Default: **0.35** | Range: 0.05 to 0.95 | decimal

What it does: How large the base QR must appear in the camera frame before the robot decides it has 'arrived' and ends the mission. It is the fraction of the whole frame the QR box fills.

Tuning: Higher value = the robot drives closer (QR must look bigger) before stopping. Lower value = it stops farther away. If it halts too far from the dock, raise it; if it bumps the base, lower it. With an A4 print, start around 0.35 and adjust by 0.05 at a time.

Driving speeds

Motor speeds (0.0–1.0) used during autonomous navigation.

Forward speed (AUTO_SPEED_FORWARD)

Default: **0.6** | Range: 0.0 to 1.0 | decimal

What it does: Motor speed when driving forward to approach a plant or the base (0.0 = stopped, 1.0 = full speed).

Tuning: Higher = reaches the target faster but overshoots more and is harder to stop centred. Lower = gentler, more accurate approach but slower runs. 0.5–0.7 is a good range on most surfaces; drop it on slippery floors.

Reverse speed (AUTO_SPEED_BACKWARD)

Default: **0.5** | Range: 0.0 to 1.0 | decimal

What it does: Motor speed when reversing during the post-watering retreat.

Tuning: Higher = backs away quickly to find the next plant sooner. Lower = gentler. Usually a touch lower than forward speed so the robot doesn't lurch backward.

Turn speed (AUTO_SPEED_TURN)

Default: **0.75** | Range: 0.0 to 1.0 | decimal

What it does: Normal left/right turning speed when aligning with a target. (Note: a previous x2 multiplier was removed, so this is now the true speed sent to the motors.)

Tuning: Higher = aligns faster but tends to swing past the target and oscillate left-right. Lower = slower but more precise centring. If the robot keeps overshooting side to side, lower this first.

Gentle turn speed (AUTO_SPEED_TURN_GENTLE)

Default: **0.375** | Range: 0.0 to 1.0 | decimal

What it does: The slower turn speed automatically used when the robot detects it is oscillating (turning left, right, then left again past the same target).

Tuning: Should be clearly lower than the normal turn speed. If oscillation still happens even after the robot switches to gentle mode, lower this further (e.g. 0.3).

Minimum turn speed (MIN_TURN_SPEED)

Default: **0.375** | Range: 0.0 to 1.0 | decimal

What it does: A floor applied to turn speed so a commanded turn never drops so low that the motors buzz but don't actually move.

Tuning: Find the lowest speed at which your motors reliably rotate the chassis, and set this just above it. If gentle turns stall, raise this; if turns are always too fast, your gentle speed may be getting clamped up by this floor.

Movement timing

How long the robot moves per step, and how long it waits to re-check the camera (seconds).

Turn step (MOVE_TURN_DURATION)

Default: **0.5** | Range: 0.05 to 5.0 | decimal

What it does: How long (seconds) the motors run for a single left/right correction before stopping to look again.

Tuning: Higher = bigger turn per step (fewer steps, but risk overshooting). Lower = small, fine corrections (more accurate, but alignment takes more cycles). Pair a short duration with a moderate turn speed for smooth centring.

Forward step (MOVE_FORWARD_DURATION)

Default: **1.0** | Range: 0.05 to 5.0 | decimal

What it does: How long (seconds) the motors run for a single forward approach step before re-checking the camera.

Tuning: Higher = covers more ground per step (faster approach, but can blow past the watering distance). Lower = cautious, stops to re-evaluate often. Lower it if the robot tends to arrive too close or off-centre.

Look-again wait (SLEEP_WAIT_YOLO)

Default: **1.5** | Range: 0.1 to 10.0 | decimal

What it does: After stopping, how long (seconds) to wait for fresh camera detections before deciding the next move.

Tuning: Higher = gives the detector more time, more reliable on a slow Pi, but makes the robot feel sluggish. Lower = snappier reactions but may act on a stale or empty frame. If the robot 'forgets' a plant it just saw, raise this.

Scan turn (SCAN_TURN_DURATION)

Default: **1.0** | Range: 0.1 to 5.0 | decimal

What it does: How long (seconds) the robot rotates per step while scanning for a plant or the base when nothing is in view.

Tuning: Higher = sweeps a wider arc per step and finds far-off targets faster, but can spin past them. Lower = slow, careful sweep that's less likely to miss. Lower it if the robot keeps rotating past the base QR without locking on.

Watering & approach

When to water, how long, and how it retreats afterwards.

Plant arrival size (ARRIVAL_AREA_RATIO)

Default: **0.5** | Range: 0.05 to 0.95 | decimal

What it does: How large a plant must appear in the frame before the robot stops and waters it (fraction of the frame filled by the plant's box).

Tuning: Higher = robot gets closer before watering. Lower = waters from farther back. Set it to match how close the pump nozzle needs to be to the plant. If it waters too early (short of the plant), raise it.

Center zone width (CENTER_MARGIN)

Default: **0.33** | Range: 0.05 to 0.49 | decimal

What it does: The width of the middle 'CENTER' zone, as a fraction of the frame. Inside this band the target is considered centred (drive forward); outside it the robot turns.

Tuning: Higher = wider centre band, so the robot drives forward sooner and turns less (faster but less precisely aimed). Lower = stricter centring, more turning before approach. 0.33 means the middle third is 'centre'.

Watering seconds (WATERING_DURATION)

Default: **5** | Range: 1 to 60 | whole number

What it does: How many seconds the pump runs on each plant.

Tuning: Set it to how long your pump needs to deliver the right amount of water. Purely a watering-volume control; it does not affect navigation.

Retreat seconds (RETREAT_DURATION)

Default: **4.0** | Range: 0.0 to 20.0 | decimal

What it does: How many seconds the robot reverses after watering, while watching for the next unwatered plant to appear.

Tuning: Higher = backs up farther (helpful when plants are spaced far apart or to get the just-watered plant out of view). Lower = stays close. If it keeps re-detecting the plant it just watered, increase this.

Retreat check rate (RETREAT_POLL_PERIOD)

Default: **0.1** | Range: 0.02 to 2.0 | decimal

What it does: How often (seconds) the robot checks for a new plant while it is reversing.

Tuning: Lower = more responsive (spots the next plant sooner) at a tiny CPU cost. Higher = lighter load. 0.1 is fine; rarely needs changing.

Lost patience (LOST_THRESHOLD)

Default: **3** | Range: 1 to 20 | whole number

What it does: How many consecutive 'look' cycles with no detection the robot tolerates (holding still) before it gives up and starts actively scanning/rotating.

Tuning: Higher = waits patiently longer before scanning (good if detections flicker in and out). Lower = starts searching sooner (good if targets rarely reappear on their own). Combine with SLEEP_WAIT_YOLO to control how long 'holding' really lasts.

Plant memory (ReID)

Appearance capture used so already-watered plants aren't watered twice.

Capture samples (REID_CAPTURE_SAMPLES)

Default: **5** | Range: 1 to 30 | whole number

What it does: How many appearance snapshots are taken of a plant right after watering it, so the robot can recognise it later and not water it twice.

Tuning: Higher = more reliable re-recognition from different angles, but the robot stays parked at the plant a little longer. Lower = faster runs, weaker memory.

Capture interval (REID_CAPTURE_INTERVAL)

Default: **0.2** | Range: 0.05 to 2.0 | decimal

What it does: Seconds between those post-watering appearance snapshots.

Tuning: Small values capture a burst quickly; larger values spread captures over time (slightly more varied views). Total parked time is about samples x interval.

Retreat capture interval (REID_RETREAT_INTERVAL)

Default: **0.4** | Range: 0.05 to 2.0 | decimal

What it does: Seconds between extra appearance snapshots taken of the just-watered plant while the robot is reversing away from it (teaches it the far/angled look).

Tuning: Lower = more views captured during retreat (better memory, more CPU). Higher = fewer. This is what stops the robot re-approaching a plant once it's far away.

Obstacle avoidance

Ultrasonic obstacle detection and the avoidance manoeuvre.

Obstacle distance (cm) (ULTRASONIC_OBSTACLE_CM)

Default: **20** | Range: 2 to 400 | whole number

What it does: Anything detected closer than this distance (in centimetres) by the ultrasonic sensor is treated as an obstacle to avoid.

Tuning: Higher = the robot reacts to obstacles earlier / from farther away (more cautious, but may dodge things it could pass). Lower = lets it get closer before avoiding (riskier). Set it a bit larger than your stopping distance at the avoid speed.

Sensor poll interval (ULTRASONIC_POLL_INTERVAL)

Default: **0.5** | Range: 0.05 to 5.0 | decimal

What it does: How often (seconds) the sensor head is asked for a fresh distance reading when the path is clear.

Tuning: Lower = detects obstacles more responsively, slightly more serial traffic. Higher = lighter. 0.5 s is a good balance.

Scan cooldown (ULTRASONIC_SCAN_COOLDOWN)

Default: **3.0** | Range: 0.0 to 30.0 | decimal

What it does: Minimum seconds between full left/centre/right servo sweep scans (a sweep is slow because the servo has to settle at each angle).

Tuning: Higher = fewer sweeps, calmer behaviour near obstacles. Lower = re-sweeps sooner. Keep it at or above ~2 s so sweeps don't pile up.

Avoid speed (ULTRASONIC_AVOID_SPEED)

Default: **0.5** | Range: 0.0 to 1.0 | decimal

What it does: Motor speed used during the obstacle-avoidance manoeuvre (back up, turn, go around).

Tuning: Higher = manoeuvres quickly but less precisely near the obstacle. Lower = safer, more controlled avoidance. Usually a bit lower than the normal driving speed.

Avoid: back up (ULTRASONIC_AVOID_BACKWARD_SECONDS)

Default: **1.0** | Range: 0.0 to 10.0 | decimal

What it does: How long the robot reverses when it first hits an obstacle, to make room before turning.

Tuning: Higher = backs up more before going around (needed in tight spaces). Lower = quick nudge back. Increase if the robot clips the obstacle while turning.

Avoid: turn (ULTRASONIC_AVOID_TURN_SECONDS)

Default: **0.8** | Range: 0.0 to 10.0 | decimal

What it does: How long the robot turns away from the obstacle (used twice: turn away, then turn back to resume heading).

Tuning: Higher = turns through a larger angle (wider detour). Lower = tighter detour. Tune together with avoid speed — angle depends on both.

Avoid: go around (ULTRASONIC_AVOID_FORWARD_SECONDS)

Default: **1.2** | Range: 0.0 to 10.0 | decimal

What it does: How long the robot drives forward to get past the obstacle before turning back to its original heading.

Tuning: Higher = travels farther around the obstacle (clears wider objects). Lower = shorter detour. Increase for large obstacles.

Hardware / wiring (read-only reference)

These appear in the UI but are not editable there. Change them in components/config.py and restart the backend. Listed here so you have the full picture.

Label	Key	Default
Camera stream URL	CAMERA_SOURCE	http://169.254.138.53/axis-cgi/mjpg/video.cgi
Motor PWM frequency (Hz)	MOTOR_PWM_FREQUENCY	1000
Left motor FWD pin	MOTOR_LEFT_FWD_PIN	12
Left motor BWD pin	MOTOR_LEFT_BWD_PIN	13
Left motor EN pin	MOTOR_LEFT_EN_PIN	5
Left motor EN pin 2	MOTOR_LEFT_EN_PIN1	17
Right motor FWD pin	MOTOR_RIGHT_FWD_PIN	22
Right motor BWD pin	MOTOR_RIGHT_BWD_PIN	23
Right motor EN pin	MOTOR_RIGHT_EN_PIN	6
Right motor EN pin 2	MOTOR_RIGHT_EN_PIN1	27
Pump relay pin	PUMP_PIN	25
Pump active-high	PUMP_ACTIVE_HIGH	False
LED data pin	LED_PIN	18
LED count	LED_COUNT	8
LED brightness	LED_BRIGHTNESS	0.2
Ultrasonic serial port	ULTRASONIC_SERIAL_PORT	/dev/ttyUSB0
Ultrasonic baud	ULTRASONIC_SERIAL_BAUD	115200
Ultrasonic timeout	ULTRASONIC_SERIAL_TIMEOUT	3.0

Appendix: ultrasonic sensor & serial port

Is it connected? On backend start the log shows either 'Ultrasonic ESP32 initialised (/dev/ttyXXX)' (success, names the port) or a WARNING 'serial unavailable...' / 'did not return a distance...' for each port it tried.

Obstacle logs. When a reading crosses the threshold the log prints 'Ultrasonic front: 18.3 cm (obstacle)' and '(clear)' on the way back — only on changes, so it won't spam. In auto mode the navigator also logs 'obstacle detected; avoiding to the left/right'.

Wrong port? The driver auto-tries, in order: the port in config, then /dev/ttyACM0, then /dev/ttyUSB0 — covering the two common ESP32 USB chips. If your board enumerates as something else (e.g. /dev/ttyUSB1), set ULTRASONIC_SERIAL_PORT in components/config.py and restart.

Find the port. Run 'ls /dev/ttyUSB* /dev/ttyACM*' (unplug/replug the ESP32 and see which entry appears/disappears), or 'dmesg | grep tty' right after plugging it in.

Hardware note. The HC-SR04 ECHO pin outputs 5 V but the ESP32 GPIO is 3.3 V (not 5 V tolerant) — use a voltage divider on ECHO to protect the board.